

Numeration systems

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<https://www.youtube.com/watch?v=cZH0YnFpjwU>

- **Numeration systems** are used to represent and manipulate numeric values.
- We have to distinguish between a **numeric value** and a **numeric representation**:
 - a **value** (abstract concept) is a measure of an attribute of a collection of objects;
 - a **representation** is the means by which we display a value and manipulate it.
- A value has different representations.
- **Example**: the numeric value “5” has more representations:
 - **5** (decimal system) , **V** (Roman system), **101**(binary system)
 - **cinci** (Romanian language), **five** (English language)

Numeration systems (contd.)

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At the mental level **people work** with **values**, but for displaying and manipulating them, different numeration systems are used (now the decimal system).

Computers work with **number representations** using the binary system.

- **Numeration system** - a set of rules for the representation and manipulation (operations) of numerical values using symbols called **digits**.
- The total number of digits is called **numeration base**(radix).

Classification of numeration systems:

- ◆ **positional systems** and **non-positional systems**

Non-positional systems: Roman system (500BC)

- **digits:** I, V, X, L, C, D, M
values: 1 5 10 50 100 500 1000
- each symbol (digit) represents a value
- it is an **additive system**:
 - the value of a number representation is obtained as the sum (difference) of the values of its digits according to some rules

Disadvantages:

- the value 0 cannot be represented
- there are different representations of the same value
example: **CDXC** and **XD** have the same value **490**
- the much greater difficulty of performing mathematical operations, such as addition, subtraction, multiplication, and division



Roman system - rules

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- a) the value of two or more identical consecutive digits is the sum of the values of these digits.

ex: *numeric representations:* III , CCC, MM
numeric values: 3 , 300, 2000

- b) the value of a pair of different digits, with the biggest one in front of the other is the sum of these two digits values:

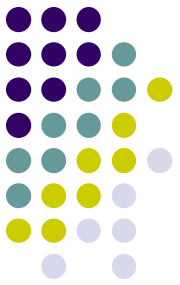
ex: *numeric representations:* VI , CL MD
numeric values: $5+1=6$, $100+50=150$, $1000+500=1500$

- c) the value of a pair of different digits, with the smallest one in front of the other is the difference of these two digits values:

ex: *numeric representations:* XC , IX , CM
numeric values: $100-10=90$, $10-1=9$, $1000-100=900$

- d) for big numbers a horizontal line over the digit is used, meaning the multiplication of the digit's value with 1000:

ex: the numeric value of $\overline{V}$ is 5000



Positional systems:

Hindu-Arabic numeration systems

- the position of a digit in a representation implies an association with a “positional value”
- the numeric value is the sum of the positional values of all the digits from the representation
 - binary system: base = 2, digits : 0,1 , **ex: 11001₍₂₎**
 - octal system: base = 8, digits: 0-7, **ex: 7564₍₈₎**
 - decimal system(1800BC) base =10, digits: 0-9, **ex: 2343**
 - hexadecimal system: base =16, digits: 0-9,A,B,C,D,E,F
where: $A_{(16)}=10$, $B_{(16)}=11$, $C_{(16)}=12$, $D_{(16)}=13$, $E_{(16)}=14$, $F_{(16)}=15$,
example: 2AF₍₁₆₎

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| | | ⊔ $1 * 2^0 = 1$ (positional value)
| | ⊔ $1 * 2^1 = 2$ (positional value)
| ⊔ $0 * 2^2 = 0$ (positional value)
⊔ $1 * 2^3 = 8$ (positional value)

- | | └─ $F_{(16)} * 16^0 = 15$ (positional value)
- | └─ $A_{(16)} * 16^1 = 10 * 16 = 160$ (positional value)
- └─ $2_{(16)} * 16^2 = 2 * 256 = 512$ (positional value)



The calculation of the *decimal values* of numbers given in different numeration bases represents a conversion method from an arbitrary base into base 10, a particular case of the *substitution method*.

Let us consider a real number in base b represented as: $(a_m a_{m-1} \dots a_1 a_0, a_{-1} \dots a_{-n})_{(b)}$.

The decimal numeric value of this representation is calculated as follows:

$$N = a'_0 * b^0 + a'_1 * b^1 + \dots + a'_m * b^m + a'_{-1} * b^{-1} + \dots + a'_{-n} * b^{-n}, \text{ where:}$$

$$a'_0 = (a_0)_{(b)}, \dots, a'_m = (a_m)_{(b)}, a'_{-1} = (a_{-1})_{(b)}, \dots, a'_{-n} = (a_{-n})_{(b)}$$

- the digits from the representation in base b are converted into decimal and then the operations $*, +, /$ are performed in base 10.
- if $b < 10$ the digits of the base b are the same in base 10.
- $a'_i * b^i$ is the positional value of the digit of rank (position) i in the representation
- the decimal numeric value of a number in a base is the sum of the positional values of all the digits from the representation in that base.



Conversion from decimal into an arbitrary base – successive divisions

Examples (F.Boian):

$$24 = 11000_{(2)}$$

24	2				
24	12	2			
0	12	6	2		
	0	6	3	2	
		0	2	1	2
			1	0	0
				1	

$$256 = 400_{(8)}$$

256	8		
256	32	8	
0	32	4	8
	0	0	0
		4	

$$2653 = A5D_{(16)}$$

2653	16		
16	165	16	
105	160	10	16
96	5	0	0
93		10	
80			
13			
0			
	5		
		A	



Example:

$543,17 = ?_{(5)}$, with 3 digits at the fractional part

- conversion of the **integer part**: successive divisions by 5 are applied.
 - the remainders are taken in the reverse order

$$543: 5 = 108 \text{ remainder } 3$$

$$108: 5 = 21 \text{ remainder } 3$$

$$21: 5 = 4 \text{ remainder } 1$$

$$4: 5 = 0 \text{ remainder } 4$$

$$543 = 4133_{(5)}$$

- conversion of the **fractional part**: successive multiplications by 5 are applied.
 - the integer parts are used in the order of obtaining them

$$0,17 * 5 = 0,85$$

$$0,85 * 5 = 4,25$$

$$0,25 * 5 = 1,25$$

$$0,25 * 5 = 1,25 \quad !!! \text{ periodicity.}$$

$$0,17 = 0,04(1)_{(5)}$$

$$543,17 = 4133,04(1)_{(5)}$$

Correspondence table – bases: 10,2,4,8,16



base 10	base $2=2^1$	base $4=2^2$	base $8=2^3$	base $16=2^4$
0	0	0	0	0
1	1	1	1	1
2	10	2	2	2
3	11	3	3	3
4	100	10	4	4
5	101	11	5	5
6	110	12	6	6
7	111	13	7	7
8	1000	20	10	8
9	1001	21	11	9
10	1010	22	12	A
11	1011	23	13	B
12	1100	30	14	C
13	1101	31	15	D
14	1110	32	16	E
15	1111	33	17	F



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Rapid conversions:

conversions between bases: 2,4,8,16



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1. *Conversion from the source base $p=2^k$, $p \in \{4=2^2, 8=2^3, 16=2^4\}$ into the destination base 2*

Rule:

Each digit from the source number in base $p=2^k$, the integer part and the fractional one, will be replaced by the corresponding group of k binary digits (adding if it is necessary insignificant zeros to the left) according to the above table .

Examples:

$$3201_{(4)} = \mathbf{11\ 10\ 00\ 01}_{(2)}$$

$$7205,346_{(8)} = \mathbf{111\ 010\ 000\ 101,011\ 100\ 110}_{(2)}$$

$$60FA,B28_{(16)} = \mathbf{0110\ 0000\ 1111\ 1010,1011\ 0010\ 1000}_{(2)}$$

Rapid conversions (contd.)

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2. Conversion from the source base 2 into the destination base $q=2^k$,
 $q \in \{4=2^2, 8=2^3, 16=2^4\}$

Rules:

- for the integer/fractional part: from *right/left* to *left/right* (relative to the decimal point) make groups of k binary digits (eventually we add to the *left/right* insignificant zeros to have a complete group);
- the groups will be replaced by the corresponding digits in base $q=2^k$ according to the above table.

Examples:

$$1\ 001\ 100\ 011\ 010\ 111_{(2)} = \boxed{001}\ \boxed{001}\ \boxed{100}\ \boxed{011}\ \boxed{010}\ \boxed{111}_{(2)} = 114327_{(8)}$$

$$\boxed{01}\ \boxed{11}\ \boxed{00}\ \boxed{00}\ \boxed{11}\ \boxed{00}\ \boxed{10}\ \boxed{10},\ \boxed{01}\ \boxed{10}\ \boxed{10}_{(2)} = 13003022,122_{(4)}$$

$$\boxed{0011}\ \boxed{0001}\ \boxed{0101}\ \boxed{1001}\ \boxed{1100}\ \boxed{1010},\ \boxed{1010}\ \boxed{1000}_{(2)} = 3159CA, A8_{(16)}$$

Conversion methods

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Conversions between arbitrary numeration bases for real numbers:

$$N_{(b)} = N'_{(h)},$$

b - source base, h - destination base

- 1. the *substitution method***
- 2. the *method of successive divisions/multiplications***
- 3. the *method that uses an intermediate base***

1. *Substitution method*: calculation performed in the destination base

Let $N_{(b)} = (a_m a_{m-1} \dots a_1 a_0, a_{-1} \dots a_{-n})_{(b)}$ be a real number in the source base b .

We want to convert the number into the destination base h .

Steps:

- all the digits from the source representation are converted into the destination base

$$(a_i)_{(b)} = (a'_i)_{(h)}, i = -n, \dots, -1, 0, \dots, m-1$$

- the base b is converted into base h : $b = (b')_{(h)}$

- we calculate in base h the following sum:

$$\begin{aligned} (N')_{(h)} = & (a'_0)_{(h)} * (b')_{(h)}^0 + (a'_1)_{(h)} * (b')_{(h)}^1 + \dots + (a'_m)_{(h)} * (b')_{(h)}^m + \\ & + (a'_{-1})_{(h)} * (b')_{(h)}^{-1} + \dots + (a'_{-n})_{(h)} * (b')_{(h)}^{-n} \end{aligned}$$

Note: The method is recommended for $b < h$, because:

$(a_i)_{(b)} = (a'_i)_{(h)}, i = -n, \dots, -1, 0, \dots, m-1$, $b = b_{(h)}$, and we have to perform only multiplications/divisions by one digit.

Substitution method - examples



Example: $354_{(6)} = ?_{(8)}$

$354_{(6)} =$

$$3_{(6)} = 3_{(8)}, 5_{(6)} = 5_{(8)}, 4_{(6)} = 4_{(8)}, 6 = 6_{(8)}$$

Calculation in base 8:

$$\begin{aligned} 354_{(6)} &= 4_{(8)} * 6_{(8)}^0 + 5_{(8)} * 6_{(8)}^1 + 3_{(8)} * 6_{(8)}^2 = 4_{(8)} + 36_{(8)} + 3_{(8)} * 44_{(8)} \\ &= 42_{(8)} + 154_{(8)} = \mathbf{216_{(8)}} \end{aligned}$$

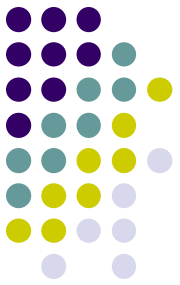
Example: $11011,11_{(2)} = ?_{(4)}$

$11011,11_{(2)} =$

$$= 2_{(4)}^4 + 2_{(4)}^3 + 2_{(4)}^1 + 2_{(4)}^0 + 2_{(4)}^{-1} + 2_{(4)}^{-2} =$$

$$= 10_{(4)}^2 + 10_{(4)} * 2_{(4)} + 2_{(4)} + 1_{(4)} + 0,2_{(4)} + 0,2_{(4)} / 2_{(4)} =$$

$$= 100_{(4)} + 20_{(4)} + 2_{(4)} + 1_{(4)} + 0,2_{(4)} + 0,1_{(4)} = \mathbf{123,3_{(4)}}$$



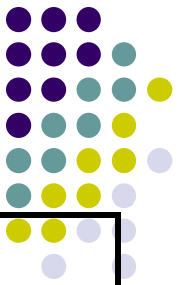
2. The method of successive divisions and multiplications

Let $N_{(b)} = (a_m a_{m-1} \dots a_1 a_0, a_{-1} \dots a_{-n})_{(b)}$ be a real number in the source base b .

We want to convert the number into the destination base h .

- it is recommended for $b > h$
- calculations are performed in the source base (b)
- for the integer part the method of **successive divisions** by the destination base (h) is applied
- for the fractional part we apply a complementary method: **successive multiplications** by the destination base (h)

The method of successive divisions

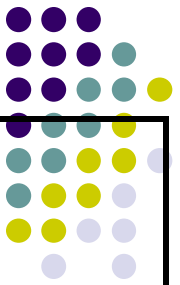


- the integer part is divided by h (destination base) obtaining a quotient and a remainder.
- the quotient is divided by h obtaining a new quotient and a new remainder...
- the process of successive divisions ends when 0 is obtained as quotient.
- the remainders, in the reverse order of obtaining them, are the digits of the new representation in base h .

$$2653 = A5D_{(16)}$$

2653	16		
16	165	16	
105	160	10	16
96	5	0	0
93		10	
80			
13			
D			
		5	
			A

The method of *successive multiplications*



- the fractional part is multiplied by h obtaining a number with an integer part and a fractional one;
- we continue with the multiplication of this new fractional part,...
- the process of the successive multiplications continues until one of the following conditions is satisfied:
 - a) the fractional part becomes 0;
 - b) an established number of digits of the fractional part were calculated
 - c) periodicity is obtained.
- the integer parts, in the order of obtaining them are the digits of the fractional part in the destination representation.

Example:

$$0,17 = ?_{(5)}$$

$$0,17 * 5 = \mathbf{0},85$$

$$0,85 * 5 = \mathbf{4},25$$

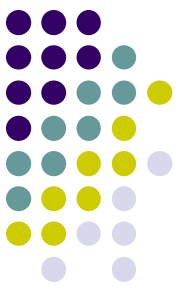
$$0,25 * 5 = \mathbf{1},25$$

$$0,25 * 5 = \mathbf{1},25$$

!!! periodicity.

$$\mathbf{0,17=0,04(1)}_{(5)} ,$$

The method of *successive divisions and multiplications*



Example: $A5B_{(16)} = ?_{(8)}$ with 3 digits at the fractional part

conversion of the integer part

$$\begin{array}{r}
 A5B_{(16)} \mid \underline{8_{(16)}} \\
 \hline
 25 \quad \mid 14B \mid \underline{8_{(16)}} \\
 \hline
 25 \quad \mid 4B \quad \mid 29 \mid \underline{8_{(16)}} \\
 \hline
 5B \quad \mid 4B \quad \mid 1 \mid 5 \mid \underline{8_{(16)}} \\
 \hline
 3 \quad \mid 3 \quad \mid 1 \quad \mid 5 \mid 0 \\
 \hline
 3 \quad \mid 3 \quad \mid 1 \quad \mid 5 \mid 0 \\
 \hline
 A5B_{(16)} = 5133_{(8)}
 \end{array}$$

conversion of the fractional part

$$0,ED_{(16)} * 8_{(16)} = 7,68_{(16)}$$

$$0,68_{(16)} * 8_{(16)} = 3,40_{(16)}$$

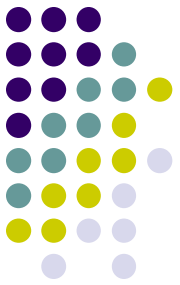
$$0,4_{(16)} * 8_{(16)} = 2,0_{(16)}$$

$$0,ED_{(16)} = 0,732_{(8)}$$

- the process of the successive multiplications is a finite one.

The final result: $A5B,ED_{(16)} = 5133,732_{(8)}$

3. The method that uses an intermediate base



$$N_{(b)} = N'_{(g)} = N''_{(h)}$$

***b* - source base, *g* – intermediate base, *h* - destination base**

a) usually $g = 10$ – calculation in base 10

- conversion from base *b* into base 10 – using the substitution method,
- conversion from base 10 into base *h* – using the method of successive divisions/multiplications

b) if *b* and *h* are powers of 2, then $g = 2$ and rapid conversions are applied

c) if $b = 10$ and $h = 2$ then for efficiency we use $g = 8$ or $g = 16$

- conversion from base 10 into base *g* - the method of successive divisions/multiplications is applied
- conversion from base *g* into base 2 – rapid conversions are applied